THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the expression $n^2 - n + 41$, the smallest natural number n for which the value is NOT prime is: 1

(a) 42 **(b)** 41 **(c)** 40 **(d)** 43

2. The sum of 7 consecutive integers is 315. The largest of them is: 1

(a) 50 **(b)** 49 **(c)** 48 **(d)** 47

3. If n is any integer not divisible by 3, then n^2 leaves a remainder ___ on division by 3. For 11 this gives: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

4. For the two-digit number 56, the value of (number − digit sum) ÷ 9 is: 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 7

5. If the sides of a triangle are 28, 45 and x (with x the longest side) and it is right-angled, then x is: 1

(a) 54 **(b)** 55 **(c)** 53 **(d)** 52

6. The smallest natural number with exactly 9 factors (counting 1 and itself) is: 1

(a) 34 **(b)** 40 **(c)** 37 **(d)** 36

7. The number of perfect squares between 400 and 3000 (both excluded) is: 1

(a) 33 **(b)** 34 **(c)** 36 **(d)** 35

8. For the expression $n^2 - n + 107$, the smallest natural number n for which the value is NOT prime is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 6

9. **Assertion (A):** The statement 'If $x = 3$, then $x^2 = 9$' is true. 1

Reason (R): The converse of every true statement is true.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** One example that satisfies a statement is enough to prove the statement true in general. 1
Reason (R): One counterexample is enough to prove a statement false.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that "If a number is divisible by 6 and by 9, then it is divisible by 54" is false. Give a counterexample. 2

12. Verify that $n^2 - 1$ is a multiple of 8 for the odd number $n = 15$. Write the converse of "If n is odd, then $n^2 - 1$ is a multiple of 8". 2

13. Use the converse of the Baudhāyana–Pythagoras theorem: for what value of x is a triangle with sides $13, 84, x$ (x the longest) right-angled? 2

14. Show that "If a number is divisible by 10 and by 15, then it is divisible by 150" is false. Give a counterexample. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the sum of the squares of two consecutive integers is odd. Verify it for 15 and 16. 3

16. Prove: if 5 divides x and x divides y, then 5 divides y. Verify with $x = 30$ and $y = 90$. 3

17. For $m = 7$ and $n = 4$ show that $(33,\ 56,\ 65)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Prove that the sum of the squares of two consecutive integers is odd. Verify it for 19 and 20. 3

18. Prove: if 9 divides x and x divides y, then 9 divides y. Verify with $x = 18$ and $y = 126$. 3

OR

For $m = 8$ and $n = 3$ show that $(55,\ 48,\ 73)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ria claims: 'The sum of any two prime numbers is even.' Her teacher asks the class to test claims by finding counterexamples or giving proofs. 5
(i) Is Ria's claim true or false?
(ii) Give a counterexample to it.
(iii) Give a counterexample to the claim '$2n + 1$ is prime for every natural number n'.
(iv) Prove that the product of two odd numbers is odd.

20. Prove that the product of three consecutive integers is divisible by 6. Verify it for $17, 18, 19$. 5

OR

Prove that the square of any odd integer is 1 more than a multiple of 8. Verify it for 21. 5

END OF QUESTION PAPER — Total: 40 Marks